

AI / Data Science

Task Report

Titanic Dataset · Exploratory Data Analysis · 3 Visualizations

DATE

12 June 2026

DOMAIN

Artificial Intelligence / Data Science

DATASET

Titanic Passenger Dataset (891 rows × 12 cols)

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Task Overview

Day 01 – Internship Training Program

Objective

The objective of Day 01 of the internship training program is to familiarise participants with their respective domains and provide hands-on experience with fundamental tools and workflows. For the AI / Data Science domain, participants are required to work with a real-world dataset and demonstrate their ability to explore, clean, and visualise data meaningfully.

Requirements

- ✓ **Analyse a given CSV dataset**
Load and inspect a real CSV file using Python and the pandas library.
- ✓ **Perform basic data exploration and preprocessing**
Understand the data structure, identify missing values, and clean the dataset.
- ✓ **Create and present 3 meaningful visualisations with insights**
Produce 3 different types of charts, each answering a clear question about the data.

Tools and Technologies Used

Tool / Library	Version	Purpose
Python	3.11+	Core programming language
pandas	latest	Data loading, exploration, and preprocessing
numpy	latest	Numerical operations and array handling
matplotlib	latest	Core chart drawing and figure management
seaborn	latest	Statistical visualisations with clean styling
VS Code	latest	Code editor used for development

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Dataset Description

Titanic Passenger Dataset

About the Dataset

The Titanic dataset is one of the most well-known beginner datasets in data science. It contains information about the 891 passengers aboard the RMS Titanic, which sank on 15 April 1912 after colliding with an iceberg. The dataset includes demographic details, ticket information, and — critically — whether each passenger survived or not. This makes it ideal for exploring patterns in survival rates based on factors such as gender, age, and passenger class.

Download

<https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv>

Size: 891 rows × 12 columns **Missing Values:** Age (177), Cabin (687), Embarked (2)

Source:

Column Reference

Column	Data Type	Description	Missing?
PassengerId	Integer	Unique ID for each passenger	No
Survived	Integer	Survival status: 0 = No, 1 = Yes	No
Pclass	Integer	Ticket class: 1 = First, 2 = Second, 3 = Third	No
Name	String	Full name of the passenger	No
Sex	String	Gender of the passenger (male / female)	No
Age	Float	Age of the passenger in years	Yes (177)
SibSp	Integer	Number of siblings / spouses aboard	No
Parch	Integer	Number of parents / children aboard	No
Ticket	String	Ticket number	No
Fare	Float	Passenger fare in British pounds	No
Cabin	String	Cabin number	Yes (687)
Embarked	String	Port of embarkation: C, Q, or S	Yes (2)

Note: The Cabin column has 687 missing values out of 891 rows (77%) — this column is dropped during preprocessing. Age has 177 missing values which are filled with the median.

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Project Setup

Installation and folder structure

Library Installation

All four required libraries are installed with a single terminal command. Python 3.11 or higher must already be installed on the system with the PATH variable configured.

```
pip install pandas matplotlib seaborn numpy
```

Verify Installation

```
python -c "import pandas, matplotlib, seaborn, numpy; print('All good!')"
# Expected output: All good!
```

Project Folder Structure

```
day01_task/
|
+-- data/
|   +-- titanic.csv          <- dataset file
|
+-- analysis.py              <- main Python code
+-- chart1_survival_by_gender.png
+-- chart2_age_distribution.png
+-- chart3_correlation_heatmap.png
+-- README.md                <- documentation
```

How to Run the Project

```
# Step 1: Navigate to your project folder
cd day01_task

# Step 2: Run the analysis script
python analysis.py

# Expected output:
# Shape: (891, 12)
# Missing values after cleaning: 0
# Chart 1 saved!
# Chart 2 saved!
# Chart 3 saved!
```

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Step 1 — Data Loading

Reading the CSV and initial inspection

Importing Libraries

All necessary libraries are imported at the top of the script. Each library serves a specific purpose in the data analysis pipeline.

```
import pandas as pd      # data loading and manipulation
import numpy as np       # numerical operations
import matplotlib.pyplot as plt # chart creation
import seaborn as sns    # enhanced chart styling
```

Loading the CSV File

```
df = pd.read_csv('data/titanic.csv')

print('Shape:', df.shape)      # number of rows and columns
print(df.head())              # first 5 rows of the dataset
print(df.columns.tolist())     # list of all column names
print(df.dtypes)              # data type of each column
```

Function	What it does
pd.read_csv()	Reads a CSV file from disk and loads it into a DataFrame
df.shape	Returns (rows, columns) — here it is (891, 12)
df.head()	Shows the first 5 rows for a quick preview of the data
df.columns	Lists all 12 column names in the dataset
df.dtypes	Shows whether each column is an integer, float, or string

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Step 2 — Data Exploration

Understanding the dataset in depth

Statistical Summary

```
print(df.describe())  
# Returns: count, mean, std, min, 25%, 50%, 75%, max  
# for every numerical column in the dataset
```

The describe() function provides a statistical summary of all numerical columns. Key observations from the output:

- Average age of passengers was approximately 29.7 years
- Average fare paid was approximately £32.20
- Only 38.4% of passengers survived (mean of Survived column = 0.384)
- Most passengers (>50%) were in 3rd class (Pclass median = 3)

Checking Missing Values

```
print(df.isnull().sum())  
  
# Output:  
# Age      177  <- needs to be filled  
# Cabin    687  <- too many missing, will be dropped  
# Embarked   2  <- only 2 missing, easy to fill  
# (all other columns: 0 missing values)
```

Exploring Categorical Columns

```
print(df['Sex'].value_counts())  
# male      577  
# female    314  
  
print(df['Pclass'].value_counts())  
# 3      491  
# 1      216  
# 2      184  
  
print(df['Survived'].value_counts())  
# 0      549  (did not survive)  
# 1      342  (survived)
```

Insight

There were significantly more male passengers (577) than female (314). 3rd class had the most passengers. Only 342 out of 891 passengers survived — a survival rate of just 38.4%.

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Step 3 — Preprocessing

Cleaning and preparing the dataset

Why Preprocessing is Necessary

Raw datasets almost always contain imperfections — missing values, irrelevant columns, or text values that need to be converted to numbers. Preprocessing ensures the data is clean and ready for analysis and visualisation. Skipping this step would cause errors or produce incorrect results.

Complete Preprocessing Code

```
# 1. Fill missing Age values with the median age
df['Age'] = df['Age'].fillna(df['Age'].median())

# 2. Fill missing Embarked with most common value
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])

# 3. Drop the Cabin column (77% missing – not usable)
df = df.drop(columns=['Cabin'])

# 4. Encode Sex as numbers: male=0, female=1
df['Sex_num'] = df['Sex'].map({'male': 0, 'female': 1})

# 5. Confirm no missing values remain
print('Missing values after cleaning:')
print(df.isnull().sum())
# All values should now show 0
```

Preprocessing Steps Explained

1 Fill Age with median

177 rows have no age value. We fill them with the median age (28.0) rather than the mean because the median is less affected by extreme outliers such as very young or very old passengers.

2 Fill Embarked with mode

Only 2 rows are missing the Embarked value. The mode (most common value) is 'S' (Southampton). Using the most common value for just 2 rows is a safe and simple fix.

3 Drop Cabin column

687 out of 891 rows have no cabin value — that is 77% of the data. A column with this many missing values cannot be reliably used for analysis, so it is removed entirely.

4 **Encode Sex as numbers**

Machine learning models and correlation calculations require numerical input. We create a new column `Sex_num` where male = 0 and female = 1, while keeping the original `Sex` column for readable chart labels.

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Step 4 — Visualisations

3 charts with code and insights

All visualisations use seaborn's whitegrid theme for clean, readable charts. Each chart is saved as a high-resolution PNG file (150 DPI) in the project folder.

Chart 1 — Survival Count by Gender (Bar Chart)

Question answered: Did men or women survive at a higher rate on the Titanic?

```
sns.set_theme(style='whitegrid')
plt.rcParams['figure.figsize'] = (8, 5)

plt.figure()
sns.countplot(data=df, x='Sex', hue='Survived',
              palette=['#E74C3C', '#2ECC71'])

plt.title('Survival Count by Gender')
plt.xlabel('Gender')
plt.ylabel('Number of Passengers')
plt.legend(labels=['Did not survive', 'Survived'])
plt.tight_layout()
plt.savefig('chart1_survival_by_gender.png', dpi=150)
plt.show()
print('Chart 1 saved!')
```

Insight Women had a dramatically higher survival rate than men. Approximately 74% of female passengers survived compared to only 18% of male passengers. This reflects the 'women and children first' evacuation policy followed during the disaster.

Chart 2 — Age Distribution of Passengers (Histogram)

Question answered: What age groups were most common among Titanic passengers?

```
plt.figure()
plt.hist(df['Age'], bins=20, color='steelblue', edgecolor='white')

plt.title('Age Distribution of Passengers')
plt.xlabel('Age')
plt.ylabel('Number of Passengers')
plt.tight_layout()
plt.savefig('chart2_age_distribution.png', dpi=150)
plt.show()
print('Chart 2 saved!')
```

Insight The age distribution shows a right-skewed pattern. The majority of passengers were between 20 and 40 years old. There is a small but noticeable peak near age 0–5, representing young children. Very few passengers were older than 60.

Chart 3 — Correlation Heatmap (Heatmap)

Question answered: Which numeric features are most strongly related to each other and to survival?

```
plt.figure()
numeric_cols = df[['Survived', 'Pclass', 'Age', 'SibSp', 'Parch', 'Fare', 'Sex_num']]
correlation = numeric_cols.corr()

sns.heatmap(correlation, annot=True, fmt='.2f',
            cmap='coolwarm', linewidths=0.5)

plt.title('Correlation Heatmap')
plt.tight_layout()
plt.savefig('chart3_correlation_heatmap.png', dpi=150)
plt.show()
print('Chart 3 saved!')
```

Insight The heatmap reveals that Sex_num (0.54) has the strongest positive correlation with survival — confirming women were more likely to survive. Pclass (-0.34) shows a negative correlation — higher class number (lower class) means lower survival odds. Fare (0.26) also correlates with survival, as higher fares often meant better access to lifeboats. Age has a slight negative correlation (-0.07).

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Key Findings

Summary of all insights discovered

1

Gender was the strongest survival factor

Women survived at 74% vs only 18% for men. The 'women and children first' policy had a very strong and measurable effect visible in the data.

2

Passenger class significantly affected survival

1st class passengers had much higher survival rates than 3rd class. Higher ticket prices correlated with better cabin locations and lifeboat access.

3

Most passengers were young adults aged 20–40

The age distribution shows the Titanic carried mostly working-age adults, with a smaller proportion of children and elderly passengers.

4

77% of Cabin data was missing

The Cabin column was so incomplete it had to be removed entirely. This is a common real-world data problem — not all data is usable as-is.

5

Fare correlates with survival

Passengers who paid higher fares (typically 1st class) had better survival odds, likely due to their cabin proximity to the deck and lifeboats.

6

Only 38.4% of all passengers survived

Of 891 passengers, only 342 survived. The disaster claimed 549 lives — over 61% of all passengers on board.

What Was Accomplished

This Day 01 task successfully covered the complete foundational workflow of a data science project — from loading a raw dataset all the way to producing meaningful visual insights. The Titanic dataset proved to be an excellent choice for a beginner task because it is well-structured, has a mix of numerical and categorical columns, contains real missing value challenges, and produces visually compelling and historically meaningful insights.

Skills Demonstrated

Skill	How it was applied
Data Loading	Used pandas <code>read_csv()</code> to load an external CSV file into a DataFrame
Data Exploration	Used <code>shape</code> , <code>head</code> , <code>describe</code> , <code>dtypes</code> , and <code>value_counts</code> for full inspection
Missing Value Analysis	Identified 3 columns with missing data and handled each differently
Data Preprocessing	Applied <code>fillna</code> (median and mode) and column dropping appropriately
Feature Engineering	Created a new numerical column (<code>Sex_num</code>) from a categorical one
Bar Chart	Used seaborn <code>countplot</code> to compare survival counts by gender
Histogram	Used matplotlib <code>hist</code> to visualise the age distribution
Heatmap	Used seaborn heatmap with correlation matrix to show feature relationships
File Output	Saved all 3 charts as PNG files and documented the project in <code>README.md</code>

Task Status: Complete

All three requirements were fully met: the dataset was analysed, preprocessing was performed, and 3 meaningful visualisations with written insights were produced and saved. The project is structured, documented, and ready for submission.